

Differentiable convex functions

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In these lectures, we will presentation some further characterizations of convex functions whenever there is Fréchet differentiability. This gives us another way of showing that a given function is convex.

Theorem 0.1 *Assume that f has continuous first order partial derivatives on a convex set C in $\mathbb{R}^n$. Then*

(i) *f is convex if and only if*

$$f(y) \geq f(x) + \nabla f(x) \cdot (y - x) \quad \text{for all } x, y \in C ; \quad (1)$$

(ii) *f is strictly convex on C if and only if*

$$f(y) > f(x) + \nabla f(x) \cdot (y - x) \quad \text{for all } y, x \in C \text{ with } x \neq y . \quad (2)$$

Proof: (i) Assume that f is convex on C . Let $x, y \in C$ and $0 < \lambda \leq 1$. From

$$f(x + \lambda(y - x)) = f(\lambda y + (1 - \lambda)x) \leq \lambda f(y) + (1 - \lambda)f(x) ,$$

it follows that

$$\frac{f(x + \lambda(y - x)) - f(x)}{\lambda} \leq f(y) - f(x) .$$

letting $\lambda \rightarrow 0^+$ and using the definition of the directional derivative, we get

$$\nabla f(x) \cdot (y - x) \leq f(y) - f(x)$$

which proves (6).

Conversely, assume that (6) holds and let us prove that

$$f(\lambda u + (1 - \lambda)v) \leq \lambda f(u) + (1 - \lambda)f(v) \quad \text{for all } u, v \in C \text{ and } \lambda \in [0, 1] . \quad (3)$$

Set $w = \lambda u + (1 - \lambda)v$ with $0 < \lambda < 1$. Hence we have that

$$v - w = -\frac{\lambda}{1 - \lambda}(u - w) .$$

Hence applying (6) to the vectors u, w and v, w we get that

$$f(w) + \nabla f(w) \cdot (u - w) \leq f(u) \quad (4)$$

$$f(w) - \frac{\lambda}{1-\lambda} \nabla f(w) \cdot (u - w) \leq f(v) \quad (5)$$

Now multiplying (4) and (5) respectively by λ and $1 - \lambda$ and summing we get (3).

(ii) Now assume that f is strictly convex. Let $x, y \in C$ with $x \neq y$ and $\lambda \in (0, 1]$. We have that

$$\nabla f(x) \cdot (y - x) \leq \frac{f(x + \lambda(y - x)) - f(x)}{\lambda} = \frac{f(\lambda y + (1 - \lambda)x) - f(x)}{\lambda} < f(y) - f(x)$$

where the first inequality follows from (6) and the last one from strict convexity. Hence also (7) holds.

Assume that (7) holds and let us prove that f is strictly convex. This follows similarly from the second part of the proof of (i). From definition of w , it is clear that if $u \neq v$ then we get $w \neq u$ and $w \neq v$. Therefore, (4) and (5) will hold with strict inequalities. This achieves the proof of the theorem. ■

Remark 0.2

- (i) Geometrically Theorem 0.1 means that a function with continuous first partial derivatives is convex if and only if its graph lies above the tangent hyperplane.
- (ii) From Theorem 0.1, one can deduce that any critical point x_0 of a differentiable convex function f on a convex set C is a global minimizer for f . In fact, for every $x \in C$ it follows that

$$f(x) \geq f(x_0) + \nabla f(x_0) \cdot (x - x_0) = f(x_0) \quad \text{since} \quad \nabla f(x_0) = 0.$$

Hence x_0 is a global minimizer of f on C .

In infinite dimensional normed spaces, Theorem 0.1 is formulated as follows.

Theorem 0.3 *Let $(V, \|\cdot\|)$ be normed space over the field $\mathbb{K} := \mathbb{R}$ or $\mathbb{C}$. Let C be a convex open subset of V and let $f : \Omega \rightarrow \mathbb{R}$ be a Fréchet differentiable function. Then*

- (i) f is convex if and only if

$$f(y) \geq f(x) + Df(x)(y - x) \quad \text{for all } x, y \in C; \quad (6)$$

- (ii) f is strictly convex on C if and only if

$$f(y) > f(x) + Df(x)(y - x) \quad \text{for all } y, x \in C \text{ with } x \neq y. \quad (7)$$

Remark 0.4 Notice that the gradient $\nabla f(x) \cdot (y - x)$ (scalar product) in theorem 0.1 is replaced in Theorem 0.3 by $Df(x)(y - x)$ the value of the Fréchet derivative $Df(x)$ at the vector $y - x$.

There are also some sufficient second order conditions from which we can derive the convexity or the strict convexity of a given function.

Definition 0.5 We say a square symmetric $n \times n$ matrix A is positive semi-definite (resp. positive definite) if

$$Au \cdot u \geq 0 \quad \forall u \in \mathbb{R}^n \quad (\text{resp. } Au \cdot u > 0 \quad \forall u \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}) .$$

Similarly, A is negative semi-definite (resp. negative definite) if

$$Au \cdot u \leq 0 \quad \forall u \in \mathbb{R}^n \quad (\text{resp. } Au \cdot u < 0 \quad \forall u \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}) .$$

Theorem 0.6 A square symmetric $n \times n$ matrix A is positive semi-definite (resp. positive definite) if all its eigenvalues are positive (resp. strictly positive).

In fact, assume that A is positive semi-definite and let λ be an eigenvalue of A and $v \neq 0_{\mathbb{R}^n}$ be a corresponding eigenvector. Then $Av = \lambda v$. So,

$$0 \leq Av \cdot v = (\lambda v) \cdot v = \lambda(v \cdot v) = \lambda \|v\|_E^2 \Rightarrow \lambda \geq 0 .$$

Similarly if A is positive definite, then

$$0 < Av \cdot v = (\lambda v) \cdot v = \lambda(v \cdot v) = \lambda \|v\|_E^2 \Rightarrow \lambda > 0 .$$

We recall that $\|\cdot\|_E$ denotes the Euclidean norm on $\mathbb{R}^n$.

Exercise 0.7 Let A be a positive semi-definite $n \times n$ matrix. Let $\lambda_{\min}$ and $\lambda_{\max}$ the smallest and largest eigenvalues of the matrix A , respectively. Then show that

$$\lambda_{\min} = \min_{\|v\|_E=1} Av \cdot v \quad \text{and} \quad \lambda_{\max} = \max_{\|v\|_E=1} Av \cdot v .$$

Hint: You will need to show that $\min_{\|v\|_E=1} Av \cdot v$ and $\max_{\|v\|_E=1} Av \cdot v$ are eigenvalues of the matrix A .

A. Lagrange multiplier!!!

Exercise 0.8 Let A be a square matrix $n \times n$.

- Show that $u \cdot Av = v \cdot A^T u$ for every $u, v \in \mathbb{R}^n$, where A^T denotes the transpose matrix of A .
- Deduce that $u \cdot Au = u \cdot \text{Sym}(A)u$ for every $u \in \mathbb{R}^n$ where $\text{Sym}(A) = \frac{1}{2}(A + A^T)$ is called the symmetric matrix of A .

The following theorem gives a characterization of convex functions from their hessian matrices.

Theorem 0.9 If f has continuous partial derivatives on some open convex set C in $\mathbb{R}^n$. Then if the hessian $Hf(x)$ of f is positive semi-definite (resp. positive definite) for every $x \in C$, then f is convex (resp. strictly convex) on C .

Proof: Assume that for every $x \in C$, the Hessian matrix $Hf(x)$ is positive semi-definite. It is enough to prove that the inequality (6) in Theorem 0.1 holds. Let $x, y \in C$. From Taylor's formula, there exists a vector $z \in [x, y]$ such that

$$f(y) = f(x) + \nabla f(x) \cdot (y - x) + \frac{1}{2} Hf(z)(y - x) \cdot (y - x) \geq f(x) + \nabla f(x) \cdot (y - x)$$

since $Hf(z)(y - x) \cdot (y - x) \geq 0$.

The second part of the theorem is obtained similarly by proving that the inequality (7) holds if Hf is positive definite on C . $\square$

Example 0.10

- (i) The functions: e^x on $(-\infty, \infty)$, $-\ln x$ on $(0, \infty)$, x^p on $[0, \infty)$ if $p > 1$, $-x^p$ on $[0, \infty)$ if $0 < p < 1$, $x \ln x$ on $(0, \infty)$ are strictly convex functions.
- (ii) The function $f(x, y) = 3x^2 + 2xy + y^2 - 2x + 3y - 5$ on $\mathbb{R}^2$ is strictly convex. In fact,

$$\nabla f(x, y) = (6x + 2y - 2, 2x + 2y + 3) \quad \text{and} \quad Hf(x, y) = \begin{bmatrix} 6 & 2 \\ 2 & 2 \end{bmatrix}.$$

Hence

$$(u, v) \cdot Hf(x, y)(u, v) = 2[2u^2 + (u + v)^2] > 0 \quad \text{for all } (u, v) \in \mathbb{R}^2 \text{ with } (u, v) \neq (0, 0).$$

Therefore, the Hessian matrix $Hf(x, y)$ is positive definite for every $(x, y) \in \mathbb{R}^2$ and hence f is strictly convex by Theorem 0.9.

Remark 0.11 The converse of Theorem 0.9 holds in the case of convexity. In fact, if f has continuous partial derivatives on some open convex set C in $\mathbb{R}^n$ and there exists a vector $z \in C$ such that $Hf(z)$ is not positive semi-definite, that is there is also a second nonzero vector $w \in \mathbb{R}^n$ such that $Hf(z)w \cdot w < 0$, then it is not difficult to prove that f is not convex on C .

Note however that the converse of Theorem 0.9 does not hold for strictly convex functions. In fact the functions $f(x) = x^4$ is strictly convex on $\mathbb{R}$ (one can easily prove this using the fact x^2 is strictly convex) while its second derivative $f''(x) = 12x^2$ does not satisfy the positive definiteness condition at $x = 0$.

Theorem 0.9 gives a nice characterization of convex when the partial second derivatives are continuous. However, it is sometime difficult to verify that the hessian is positive. Therefore, it might be useful to deduce convexity as a result of combination of elementary convex functions.

Theorem 0.12

- (i) If $f_1, f_2, \dots, f_k$ are convex functions defined on the convex set in $\mathbb{R}^n$. Then the function

$$f(\mathbf{x}) = \sum_{i=1}^k f_i(\mathbf{x}) \text{ is convex.}$$

Furthermore, if at least one f_i is strictly convex for some i , then f is also strictly convex.

- (ii) If f is convex (resp. strictly convex) on C and $\alpha > 0$, then αf is convex (resp. strictly convex).
- (iii) If f is convex (resp. strictly convex) and g is increasing (resp. strictly increasing) and convex on the range of f in $\mathbb{R}$, then $g \circ f$ is convex (resp. strictly convex).
- (iv) Let I be a set of indices and $(f_i)_{i \in I}$ be a family of convex functions on the convex set C in $\mathbb{R}^n$. Then the function $f(x) = \sup_{i \in I} f_i(x)$ is also convex on C .

Example 0.13 Let $\alpha_1, \alpha_2, \dots, \alpha_n$ be fixed real numbers. The function

$$f(x) = e^{\alpha_1 x_1} + e^{\alpha_2 x_2} + \dots + e^{\alpha_n x_n} = \sum_{i=1}^n e^{\alpha_i x_i} \quad \forall x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$$

is convex on $\mathbb{R}^n$ because each function $f_i(x) = e^{\alpha_i x_i}$ is convex on $\mathbb{R}^n$. If all the α_i are different from zero, then we find for every $x \in \mathbb{R}^n$ and for every $u = (u_1, u_2, \dots, u_n) \in \mathbb{R}^n$ with $u \neq 0$ that

$$Hf(x)u \cdot u = \sum_{i=1}^n u_i^2 \alpha_i^2 e^{\alpha_i x_i} > 0$$

which by Theorem 0.9, implies that f is strictly convex on $\mathbb{R}^n$.

The next lecture will be on the implicit function theorem for real valued function of two and even more variables.